Data Sciences – CentraleSupelec
Large Scale and Distributed Optimization
Part V: Fixed point algorithms

Emilie Chouzenoux and Jean-Christophe Pesquet

Center for Visual Computing
CentraleSupelec
emilie.chouzenoux@u-pem.fr, jean-christophe@pesquet.eu

A first answer

Fixed point theorem (E. Picard, 1856-1941)



If $T : \mathcal{H} \to \mathcal{H}$ is such that

- $\widehat{x} \in \mathcal{H}$ is a fixed point of T, i.e. $\widehat{x} = T\widehat{x}$
- T is a strict contraction, i.e. there exists $\rho \in [0, 1[$ such that

$$\left(\forall (x, x') \in \mathcal{H}^2\right) \qquad \|Tx - Tx'\| \leq \rho \|x - x'\|$$

then the sequence $(x_n)_{n\in\mathbb{N}}$ defined as $(\forall n \in \mathbb{N})\ x_{n+1} = Tx_n$ with $x_0 \in \mathcal{H}$, converges to $\widehat{x}$.

A first answer

Fixed point theorem (E. Picard, 1856-1941)



If $T : \mathcal{H} \to \mathcal{H}$ is such that

- $\widehat{x} \in \mathcal{H}$ is a fixed point of T, i.e. $\widehat{x} = T\widehat{x}$
- T is a strict contraction, i.e. there exists $\rho \in [0, 1[$ such that

$$(\forall (x, x') \in \mathcal{H}^2) \qquad \|Tx - Tx'\| \leq \rho \|x - x'\|$$

then the sequence $(x_n)_{n\in\mathbb{N}}$ defined as $(\forall n \in \mathbb{N})\ x_{n+1} = Tx_n$ with $x_0 \in \mathcal{H}$, converges to $\widehat{x}$.

Proof: For all $n \in \mathbb{N}$,

$$\begin{aligned} \|x_{n+1} - \widehat{x}\| &= \|Tx_n - T\widehat{x}\| \\ &\leq \rho \|x_n - \widehat{x}\|. \end{aligned}$$

Consequently, $\|x_n - \widehat{x}\| \leq \rho^n \|x_0 - \widehat{x}\|$. Hence, we have proved that $(x_n)_{n\in\mathbb{N}}$ converges linearly to $\widehat{x}$.

Objective of this part

- Extend this theorem to more general operators
  - not necessarily *strictly* contractive
  - possibly dependent on the iteration number n
  - built from composition of simpler operators
    (*splitting techniques*).

- Apply this to solve minimization problems.
  $\rightsquigarrow$ How to relate T to the objective function f ?

Fixed point algorithms



Fixed point algorithms: convergence

Let $\mathcal{H}$ be a Hilbert space.
Let $(x_n)_{n\in\mathbb{N}}$ be a sequence in $\mathcal{H}$ and $\widehat{x} \in \mathcal{H}$.

- $(x_n)_{n\in\mathbb{N}}$ converges strongly to $\widehat{x}$ if

$$\lim_{n\to+\infty} \|x_n - \widehat{x}\| = 0.$$

  It is denoted by $x_n \to \widehat{x}$.

- $(x_n)_{n\in\mathbb{N}}$ converges weakly to $\widehat{x}$ if

$$(\forall y \in \mathcal{H}) \qquad \lim_{n\to+\infty} \langle y \mid x_n - \widehat{x}\rangle = 0.$$

  It is denoted by $x_n \rightharpoonup \widehat{x}$.

Remark: In a finite dimensional Hilbert space, strong and weak convergences are equivalent.

Fixed point algorithms: convergence

Let $(x_n)_{n\in\mathbb{N}}$ be a sequence of $\mathcal{H}$.
$(x_n)_{n\in\mathbb{N}}$ converges weakly if and only if

- $(x_n)_{n\in\mathbb{N}}$ is bounded

  and
- $(x_n)_{n\in\mathbb{N}}$ possesses at most one sequential cluster point in the weak topology.

- $\widehat{x}$ is a sequential cluster point of $(x_n)_{n\in\mathbb{N}}$ in the weak topology if there exists a subsequence $(x_{n_k})_{k\in\mathbb{N}}$ of $(x_n)_{n\in\mathbb{N}}$ that converges weakly to $\widehat{x}$.

Fixed point algorithms: convergence

Let $(x_n)_{n\in\mathbb{N}}$ be a sequence of $\mathcal{H}$.
$(x_n)_{n\in\mathbb{N}}$ converges weakly if and only if

- $(x_n)_{n\in\mathbb{N}}$ is bounded

  and

- $(x_n)_{n\in\mathbb{N}}$ possesses at most one sequential cluster point in the weak topology.

Illustration:

x_0	x_1	x_2	x_3	x_4	x_5	...
1	-1	1	-1	1	-1	...

$\rightarrow (x_n)_{n\in\mathbb{N}}$ is bounded but it has 2 sequential cluster points: -1 and 1.

$\rightarrow (x_n)_{n\in\mathbb{N}}$ does not converge.

Fixed point algorithms: convergence

Lemma 1

Let D be a nonempty subset of $\mathcal{H}$.
Let $(x_n)_{n\in\mathbb{N}}$ be a sequence in $\mathcal{H}$.
$(x_n)_{n\in\mathbb{N}}$ **weakly converges** to a point in D if

- for every $x \in D$, $(\|x_n - x\|)_{n\in\mathbb{N}}$ converges

  and

- **every** weak sequential cluster point of $(x_n)_{n\in\mathbb{N}}$ lies in D.

Fixed point algorithms: convergence

Proof:

If $(\|x_n - x\|)_{n\in\mathbb{N}}$ converges, then $(\|x_n - x\|)_{n\in\mathbb{N}}$ and thus $(x_n)_{n\in\mathbb{N}}$ are bounded.
We assume that $(x_{n_k})_{k\in\mathbb{N}}$ and $(x_{n_\ell})_{\ell\in\mathbb{N}}$ are such that $x_{n_k} \rightharpoonup \widehat{x}$ and $x_{n_\ell} \rightharpoonup \widehat{x}'$ where $(\widehat{x}, \widehat{x}') \in D^2$. For every $n \in \mathbb{N}$,

$$2\left\langle x_n \mid \widehat{x}' - \widehat{x}\right\rangle = \|x_n - \widehat{x}\|^2 - \|x_n - \widehat{x}'\|^2 - \|\widehat{x}\|^2 + \|\widehat{x}'\|^2.$$

Because $(\|x_n - \widehat{x}\|)_{n\in\mathbb{N}}$ and $(\|x_n - \widehat{x}'\|)_{n\in\mathbb{N}}$ converge, there exists $\alpha \in \mathbb{R}$ such that $\langle x_n \mid \widehat{x}' - \widehat{x}\rangle \to \alpha$ and thus $\langle x_{n_k} \mid \widehat{x}' - \widehat{x}\rangle \to \langle \widehat{x} \mid \widehat{x}' - \widehat{x}\rangle = \alpha$. Similarly, $\langle \widehat{x}' \mid \widehat{x}' - \widehat{x}\rangle = \alpha$.
Consequently, $\|\widehat{x}' - \widehat{x}\|^2 = 0 \Rightarrow \widehat{x} = \widehat{x}'$.

Fixed point algorithms: Fejér-monotone sequence

Let D be a nonempty subset of a Hilbert space $\mathcal{H}$.
Let $(x_n)_{n\in\mathbb{N}}$ be a sequence in $\mathcal{H}$.
$(x_n)_{n\in\mathbb{N}}$ is **Fejér-monotone** with respect to D if

$$(\forall x \in D)(\forall n \in \mathbb{N}) \qquad \|x_{n+1} - x\| \leq \|x_n - x\|.$$

Let $D \subset \mathcal{H}$.
Let $(x_n)_{n\in\mathbb{N}}$ be Fejér-monotone with respect to D then

- for every $x \in D$, $(\|x_n - x\|)_{n\in\mathbb{N}}$ converges,
- $(x_n)_{n\in\mathbb{N}}$ is bounded.

Fixed point algorithms: Fejér-monotone sequence

Fejér-monotone convergence

Let D be a nonempty subset of a Hilbert space $\mathcal{H}$.
Let $(x_n)_{n\in\mathbb{N}}$ be a sequence in $\mathcal{H}$.
$(x_n)_{n\in\mathbb{N}}$ converges weakly to a point in D if

- $(x_n)_{n\in\mathbb{N}}$ is Fejér-monotone with respect to D

  and

- every weak sequential cluster point of $(x_n)_{n\in\mathbb{N}}$ lies in D.

Fixed point algorithms: Fejér-monotone sequence

Fejér-monotone convergence

Let D be a nonempty subset of a Hilbert space $\mathcal{H}$.
Let $(x_n)_{n\in\mathbb{N}}$ be a sequence in $\mathcal{H}$.
$(x_n)_{n\in\mathbb{N}}$ converges weakly to a point in D if

- $(x_n)_{n\in\mathbb{N}}$ is Fejér-monotone with respect to D

  and

- every weak sequential cluster point of $(x_n)_{n\in\mathbb{N}}$ lies in D.

Lemma 2

Let C be a nonempty closed convex subset of $\mathcal{H}$.
If $(x_n)_{n\in\mathbb{N}}$ denotes a sequence in C that weakly converges to $\widehat{x}$ then $\widehat{x} \in C$.

Nonexpansive operator: definition

Let C be a nonempty set of a Hilbert space $\mathcal{H}$. Let $T\colon C \to \mathcal{H}$.
The set of fixed points of T is

$$\operatorname{Fix} T = \{x \in C \mid x = Tx\}.$$

Let $S\colon C \to 2^{\mathcal{H}}$. The set of zeros of S is

$$\operatorname{zer} S = \{x \in C \mid 0 \in Sx\}.$$

Let $C \subset \mathcal{H}$ be a nonempty set.
Let $T\colon C \to \mathcal{H}$.
T is a nonexpansive operator if $\big(\forall (x,y) \in C^2\big) \quad \|Tx - Ty\| \leq \|x - y\|$.

Nonexpansive operator: fixed point algorithm

Demiclosedness principle

Let C be a nonempty closed convex subset of a Hilbert space $\mathcal{H}$.
Let $T\colon C \to \mathcal{H}$ be a nonexpansive operator.
If $(x_n)_{n\in\mathbb{N}}$ is a sequence in C that converges weakly to $\widehat{x}$ and if $Tx_n - x_n \to 0$ then $\widehat{x} \in \operatorname{Fix} T$.

Nonexpansive operator: fixed point algorithm

Demiclosedness principle

Let C be a nonempty closed convex subset of a Hilbert space $\mathcal{H}$.
Let $T : C \to \mathcal{H}$ be a nonexpansive operator.
If $(x_n)_{n\in\mathbb{N}}$ is a sequence in C that converges weakly to $\widehat{x}$ and if $Tx_n - x_n \to 0$ then $\widehat{x} \in \operatorname{Fix} T$.

Proof:

$x_n \rightharpoonup \widehat{x} \Rightarrow \widehat{x} \in C$ and $T\widehat{x}$ defined. For every $n \in \mathbb{N}$,

$$\|x_n - T\widehat{x}\|^2 = \|x_n - \widehat{x}\|^2 + \|\widehat{x} - T\widehat{x}\|^2 + 2\langle x_n - \widehat{x} \mid \widehat{x} - T\widehat{x}\rangle$$

$$\|x_n - T\widehat{x}\|^2 = \|x_n - Tx_n\|^2 + \|Tx_n - T\widehat{x}\|^2 + 2\langle x_n - Tx_n \mid Tx_n - T\widehat{x}\rangle$$

$$\Rightarrow \|\widehat{x} - T\widehat{x}\|^2 = \|x_n - Tx_n\|^2 + \|Tx_n - T\widehat{x}\|^2 - \|x_n - \widehat{x}\|^2 + 2\langle x_n - Tx_n \mid Tx_n - T\widehat{x}\rangle - 2\langle x_n - \widehat{x} \mid \widehat{x} - T\widehat{x}\rangle$$

Nonexpansive operator: fixed point algorithm

Demiclosedness principle

Let C be a nonempty closed convex subset of a Hilbert space $\mathcal{H}$.
Let $T\colon C \to \mathcal{H}$ be a nonexpansive operator.
If $(x_n)_{n\in\mathbb{N}}$ is a sequence in C that converges weakly to $\widehat{x}$ and if $Tx_n - x_n \to 0$ then $\widehat{x} \in \operatorname{Fix} T$.

Proof:

$$\begin{aligned}\|\widehat{x} - T\widehat{x}\|^2 =& \|x_n - Tx_n\|^2 + \|Tx_n - T\widehat{x}\|^2 - \|x_n - \widehat{x}\|^2 \\ &+ 2\langle x_n - Tx_n \mid Tx_n - T\widehat{x}\rangle - 2\langle x_n - \widehat{x} \mid \widehat{x} - T\widehat{x}\rangle .\end{aligned}$$

Since T is nonexpansive, by using the Cauchy-Schwarz inequality,

$$\begin{aligned}\|\widehat{x} - T\widehat{x}\|^2 &\leq \|x_n - Tx_n\|^2 + 2\|x_n - Tx_n\|\|Tx_n - T\widehat{x}\| - 2\langle x_n - \widehat{x} \mid \widehat{x} - T\widehat{x}\rangle \\ &\leq \|x_n - Tx_n\|^2 + 2\|x_n - Tx_n\|\|x_n - \widehat{x}\| - 2\langle x_n - \widehat{x} \mid \widehat{x} - T\widehat{x}\rangle .\end{aligned}$$

$x_n \rightharpoonup \widehat{x} \Rightarrow (x_n)_{n\in\mathbb{N}}$ bounded. The result follows by taking the limit.

Nonexpansive operator: fixed point algorithm

Let C be a nonempty closed convex subset of a Hilbert space $\mathcal{H}$.
Let $T : C \to C$ be a nonexpansive operator

$$(\forall n \in \mathbb{N}) \quad x_{n+1} = Tx_n.$$

Nonexpansive operator: fixed point algorithm

Let C be a nonempty closed convex subset of a Hilbert space $\mathcal{H}$.
Let $T : C \to C$ be a nonexpansive operator

$$(\forall n \in \mathbb{N}) \quad x_{n+1} = Tx_n.$$

If $x_n - Tx_n \to 0$,

Nonexpansive operator: fixed point algorithm

Let C be a nonempty closed convex subset of a Hilbert space $\mathcal{H}$.
Let $T : C \to C$ be a nonexpansive operator such that $\operatorname{Fix} T \neq \varnothing$.
Let $x_0 \in C$,

$$(\forall n \in \mathbb{N}) \quad x_{n+1} = Tx_n.$$

If $x_n - Tx_n \to 0$, then $(x_n)_{n\in\mathbb{N}}$ converges weakly to a point in $\operatorname{Fix} T$.

Exercise :

Prove this result by showing that $(x_n)_{n\in\mathbb{N}}$ is Fejér-monotone with respect to $\operatorname{Fix} T$.

Fixed point algorithms: Fejér-monotone sequence

Krasnosel'skii-Mann algorithm

Let C be a nonempty closed convex subset of a Hilbert space $\mathcal{H}$.
Let $T\colon C \to C$ be a nonexpansive operator such that $\operatorname{Fix} T \neq \varnothing$.
Let $(\lambda_n)_{n\in\mathbb{N}}$ be a sequence in $[0,1]$ such that

$$\sum_{n\in\mathbb{N}} \lambda_n(1-\lambda_n) = +\infty.$$

Let $x_0 \in C$ and $(\forall n \in \mathbb{N})\ x_{n+1} = x_n + \lambda_n(Tx_n - x_n)$. Then,

- $(x_n)_{n\in\mathbb{N}}$ is Fejér-monotone with respect to $\operatorname{Fix} T$.
- $(Tx_n - x_n)_{n\in\mathbb{N}}$ converges strongly to 0.
- $(x_n)_{n\in\mathbb{N}}$ converges weakly to a point in $\operatorname{Fix} T$.

Typical choice: $(\forall n \in \mathbb{N})\ \lambda_n = \lambda \in]0,1[$.

Fixed point algorithms: Fejér-monotone sequence

Proof :
For every $n \in \mathbb{N}$, by convex combination, $x_n \in C$.

Fejér-monotonicity with respect to $\operatorname{Fix} T$: $(\forall x \in \operatorname{Fix} T)(\forall n \in \mathbb{N})$

$$
\begin{aligned}
&\|x_{n+1} - x\|^2 \\
=&\|x_n + \lambda_n(Tx_n - x_n) - x\|^2 \\
=&\|(1-\lambda_n)(x_n - x) + \lambda_n(Tx_n - x)\|^2 \\
=&(1-\lambda_n)^2\|x_n - x\|^2 + \lambda_n^2\|Tx_n - x\|^2 - 2\lambda_n(1-\lambda_n)\langle x - x_n \mid Tx_n - x\rangle \\
=&(1-\lambda_n)\|x_n - x\|^2 + \lambda_n\|Tx_n - x\|^2 - \lambda_n(1-\lambda_n)\|Tx_n - x + x - x_n\|^2 \\
=&(1-\lambda_n)\|x_n - x\|^2 + \lambda_n\|Tx_n - Tx\|^2 - \lambda_n(1-\lambda_n)\|Tx_n - x_n\|^2 \\
\leq&(1-\lambda_n)\|x_n - x\|^2 + \lambda_n\|x_n - x\|^2 - \lambda_n(1-\lambda_n)\|Tx_n - x_n\|^2 \\
\leq&\|x_n - x\|^2.
\end{aligned}
$$

Fixed point algorithms: Fejér-monotone sequence

Proof :
We want to prove that $Tx_n - x_n \to 0$.
We deduce from $\|x_{n+1} - x\|^2 \leq \|x_n - x\|^2 - \lambda_n(1-\lambda_n)\|Tx_n - x_n\|^2$ that

$$\sum_{n\in\mathbb{N}} \lambda_n(1-\lambda_n)\|Tx_n - x_n\|^2 \leq \|x_0 - x\|^2$$

$$\Rightarrow (\forall n \in \mathbb{N}) \qquad \inf_{k\geq n} \|Tx_k - x_k\|^2 \sum_{k=n}^{+\infty} \lambda_k(1-\lambda_k) \to 0.$$

The assumptions over the sequence $(\lambda_n)_{n\in\mathbb{N}}$ lead to
$\liminf_{n\to+\infty} \|Tx_n - x_n\| = 0$.

Fixed point algorithms: Fejér-monotone sequence

Proof :
The assumptions on $(\lambda_n)_{n\in\mathbb{N}}$ lead to $\liminf_{n\to+\infty} \|Tx_n - x_n\| = 0$.
Moreover, as T is a nonexpansive operator

$$\begin{aligned}\|Tx_{n+1} - x_{n+1}\| &= \|Tx_{n+1} - Tx_n + (1-\lambda_n)(Tx_n - x_n)\| \\ &\leq \|x_{n+1} - x_n\| + (1-\lambda_n)\|Tx_n - x_n\| \\ &= \|Tx_n - x_n\|.\end{aligned}$$

Consequently, $(\|Tx_n - x_n\|)_{n\in\mathbb{N}}$ converges and

$$Tx_n - x_n \to 0.$$

Fixed point algorithms: Fejér-monotone sequence

Proof :
Let $(x_{n_k})_{k\in\mathbb{N}}$ be a subsequence of $(x_n)_{n\in\mathbb{N}}$ such that $x_{n_k} \rightharpoonup \widehat{x}$. According to the demiclosedness principle, $Tx_{n_k} - x_{n_k} \to 0$, yields $\widehat{x} \in \mathrm{Fix}\, T$. The weak convergence of $(x_n)_{n\in\mathbb{N}}$ to $\widehat{x}$ results from the Fejér-monotonicity of $(x_n)_{n\in\mathbb{N}}$ with respect to $\mathrm{Fix}\, T$.

α-averaged operator: definition

Let $C \subset \mathcal{H}$ be a nonempty set of a Hilbert space $\mathcal{H}$.
Let $A : C \to \mathcal{H}$ and let $\alpha \in]0,1[$.
A is an α-averaged operator if there exists a nonexpansive operator $R : C \to \mathcal{H}$ such that

$$A = (1-\alpha)\mathrm{Id} + \alpha R.$$

Let $C \subset \mathcal{H}$ be a nonempty set.
Let $A : C \to \mathcal{H}$ and let $\alpha \in]0,1[$.
A is an α-averaged operator if and only if

$$(\forall (x,y) \in C^2) \quad \|Ax - Ay\|^2 + \frac{1-\alpha}{\alpha}\|(\mathrm{Id}-A)x - (\mathrm{Id}-A)y\|^2 \leq \|x-y\|^2.$$

Remark : When $\alpha = 1/2$, we say that A is firmly nonexpansive.

α-averaged operator: properties

Let $C \subset \mathcal{H}$ be a nonempty set. Let $A_1 \colon C \to \mathcal{H}$ be α_1-averaged and let $A_2 \colon C \to \mathcal{H}$ be α_2-averaged with $(\alpha_1, \alpha_2) \in]0,1[^2$.

- Let $\lambda \in]0,1[$. $\lambda A_1 + (1-\lambda)A_2$ is γ-averaged with

$$\gamma = \lambda\alpha_1 + (1-\lambda)\alpha_2.$$

- Assume that $A_2(C) \subset C$. Then, $A_1 \circ A_2$ is γ-averaged with

$$\gamma = \frac{\alpha_1 + \alpha_2 - 2\alpha_1\alpha_2}{1 - \alpha_1\alpha_2}.$$

Exercise: Prove the first property.

α-averaged operator: example

Let $f \in \Gamma_0(\mathcal{H})$, $\nu \in]0, 2[$.
If f is differentiable with a ν-Lipschitzian gradient then, $\mathrm{Id} - \nabla f$ is a $\nu/2$-averaged operator.

α-averaged operator: example

Proof : 1) Descent lemma

For every $(x, y) \in \mathcal{H}^2$ and $t \in \mathbb{R}$, let $\varphi(t) = f(x + t(y - x))$.
φ is differentiable and $\varphi'(t) = \langle y - x \mid \nabla f(x + t(y - x)) \rangle$. We have then

$$\varphi(1) - \varphi(0) = \int_0^1 \varphi'(t) dt$$

$$\Leftrightarrow \; f(y) - f(x) - \langle y - x \mid \nabla f(x) \rangle = \int_0^1 \langle y - x \mid \nabla f(x + t(y - x)) - \nabla f(x) \rangle \, dt.$$

In addition, according to the Cauchy-Schwarz inequality,

$$\begin{aligned} &\langle y - x \mid \nabla f(x + t(y - x)) - \nabla f(x) \rangle \\ &\leq \|y - x\| \|\nabla f(x + t(y - x)) - \nabla f(x)\| \leq t\nu \|y - x\|^2. \end{aligned}$$

This leads to

$$\boxed{\left(\forall (x, y) \in \mathcal{H}^2\right) \quad f(y) \leq f(x) + \langle y - x \mid \nabla f(x) \rangle + \frac{\nu}{2} \|y - x\|^2.}$$

α-averaged operator: example

Proof : 2) $\mathrm{Id} - \nabla f$ is α-averaged

From the descent lemma, for every $(x, y, z) \in \mathcal{H}^3$,

$$\begin{aligned} f^*(\nabla f(y)) &\geq \langle z \mid \nabla f(y) \rangle - f(z) \\ &\geq \langle z \mid \nabla f(y) - \nabla f(x) \rangle + \langle x \mid \nabla f(x) \rangle - f(x) - \frac{\nu}{2}\|z - x\|^2. \end{aligned}$$

Moreover, according to the Fenchel-Young inequality,

$$\langle x \mid \nabla f(x) \rangle - f(x) = f^*(\nabla f(x)).$$

Thus,

$$f^*(\nabla f(y)) \geq \langle z \mid \nabla f(y) - \nabla f(x) \rangle + f^*(\nabla f(x)) - \frac{\nu}{2}\|z - x\|^2$$

α-averaged operator: example

Proof : 2) $\mathrm{Id} - \nabla f$ is α-averaged

Thus,

$$f^*(\nabla f(y)) \geq \langle z \mid \nabla f(y) - \nabla f(x)\rangle + f^*(\nabla f(x)) - \frac{\nu}{2}\|z-x\|^2$$

$$= f^*(\nabla f(x)) + \langle x \mid \nabla f(y) - \nabla f(x)\rangle + \langle z-x \mid \nabla f(y) - \nabla f(x)\rangle - \frac{\nu}{2}\|z-x\|^2.$$

This yields

$$\begin{aligned} f^*(\nabla f(y)) \geq & f^*(\nabla f(x)) + \langle x \mid \nabla f(y) - \nabla f(x)\rangle \\ & + \left(\nu\|\cdot\|^2/2\right)^*\left(\nabla f(y) - \nabla f(x)\right) \\ \geq & f^*(\nabla f(x)) + \langle x \mid \nabla f(y) - \nabla f(x)\rangle + \frac{1}{2\nu}\|\nabla f(y) - \nabla f(x)\|^2. \end{aligned}$$

Consequently,

$$\boxed{f^*(\nabla f(y)) \geq f^*(\nabla f(x)) + \langle x \mid \nabla f(y) - \nabla f(x)\rangle + \frac{1}{2\nu}\|\nabla f(y) - \nabla f(x)\|^2.}$$

α-averaged operator: example

Proof : 2) $\mathrm{Id} - \nabla f$ is α-averaged

For every $(x, y) \in \mathcal{H}^2$,

$$f^*(\nabla f(y)) \geq f^*(\nabla f(x)) + \langle x \mid \nabla f(y) - \nabla f(x) \rangle + \frac{1}{2\nu} \|\nabla f(y) - \nabla f(x)\|^2$$

and symmetrically

$$f^*(\nabla f(x)) \geq f^*(\nabla f(y)) + \langle y \mid \nabla f(x) - \nabla f(y) \rangle + \frac{1}{2\nu} \|\nabla f(x) - \nabla f(y)\|^2.$$

By summing (Baillon-Haddad theorem)

$$-\langle y - x \mid \nabla f(y) - \nabla f(x) \rangle + \frac{1}{\nu} \|\nabla f(x) - \nabla f(y)\|^2 \leq 0.$$

It results that

$$\boxed{\|(\mathrm{Id} - \nabla f)x - (\mathrm{Id} - \nabla f)y\|^2 + \frac{1 - \nu/2}{\nu/2} \|\nabla f(x) - \nabla f(y)\|^2 \leq \|x - y\|^2.}$$

α-averaged operator: example

Let $f \in \Gamma_0(\mathcal{H})$, $\nu \in \ ]0, +\infty[$ and $\gamma \in]0, 2/\nu[$.
If f is differentiable with a ν-Lipschitzian gradient then $\mathrm{Id} - \gamma\nabla f$ is a $\gamma\nu/2$-averaged operator.

Remark : $\mathrm{Id} - \gamma\nabla f$ is the gradient descent operator.

α-averaged operator: example

Let $f \in \Gamma_0(\mathcal{H})$.
prox_f is a firmly nonexpansive (i.e. $1/2$-averaged operator).

α-averaged operator: example

Proof :
Let $u_1 \in \partial f(x_1)$ and $u_2 \in \partial f(x_2)$. By monotonicity of ∂f,

$$\langle x_1 - x_2 \,|\, u_1 - u_2 \rangle \geq 0 \Leftrightarrow \quad \langle x_1 - x_2 \,|\, x_1 - x_2 + u_1 - u_2 \rangle \geq \| x_1 - x_2 \|^2.$$

If we consider $u_1' \in (\mathrm{Id} + \partial f) x_1$ et $u_2' \in (\mathrm{Id} + \partial f) x_2$, it results that

$$\langle x_1 - x_2 \,|\, u_1' - u_2' \rangle \geq \| x_1 - x_2 \|^2$$

Then, from the definition of the proximity operator,

$$\langle \mathrm{prox}_f u_1' - \mathrm{prox}_f u_2' \,|\, u_1' - u_2' \rangle \geq \| \mathrm{prox}_f u_1' - \mathrm{prox}_f u_2' \|^2$$

We can deduce that prox_f is a $1/2$-averaged operator, i.e,

$$\| u_1' - u_2' \|^2 \geq \| \mathrm{prox}_f u_1' - \mathrm{prox}_f u_2' \|^2 + \| (\mathrm{Id} - \mathrm{prox}_f) u_1' - (\mathrm{Id} - \mathrm{prox}_f) u_2' \|^2.$$

Fixed point algorithms: α-averaged operator

Let $T : \mathcal{H} \to \mathcal{H}$ be an α-averaged operator with $\alpha \in]0, 1[$ such that $\operatorname{Fix} T \neq \varnothing$.
Let $(\lambda_n)_{n\in\mathbb{N}}$ be a sequence in $[0, 1/\alpha]$ such that

$$\sum_{n\in\mathbb{N}} \lambda_n(1 - \alpha\lambda_n) = +\infty.$$

Let $x_0 \in \mathcal{H}$ and $(\forall n \in \mathbb{N})\ x_{n+1} = x_n + \lambda_n(Tx_n - x_n)$. The following properties are satisfied

- $(x_n)_{n\in\mathbb{N}}$ is Fejér-monotone with respect to $\operatorname{Fix} T$.
- $(Tx_n - x_n)_{n\in\mathbb{N}}$ converges strongly to 0.
- $(x_n)_{n\in\mathbb{N}}$ converges weakly to a point in $\operatorname{Fix} T$.

Fixed point algorithms: α-averaged operator

Proof :
Since T is α-averaged, there exists a non expansive operator R such that $T = (1-\alpha)\mathrm{Id} + \alpha R$.
Let $(\forall n \in \mathbb{N})\ \mu_n = \alpha\lambda_n \in [0,1]$.
The iterations can be written as

$$(\forall n \in \mathbb{N}) \qquad \begin{aligned} x_{n+1} &= x_n + \lambda_n(Tx_n - x_n) \\ &= x_n + \mu_n(Rx_n - x_n). \end{aligned}$$

Moreover, $\mathrm{Fix}R = \mathrm{Fix}\,T$.
$+$ Krasnosel'skii-Mann algorithm.

Optimization algorithms: *Forward-Backward*

Let $f \in \Gamma_0(\mathcal{H})$.
Let $g \in \Gamma_0(\mathcal{H})$ be differentiable with a ν-Lipschitzian gradient where $\nu \in \,]0, +\infty[$.
Let $\gamma \in]0, 2/\nu[$ and $\delta = 2 - \gamma\nu/2 \in]1, 2[$.
Let $(\lambda_n)_{n\in\mathbb{N}}$ be a sequence in $[0, \delta]$ such that $\sum_{n\in\mathbb{N}} \lambda_n(\delta - \lambda_n) = +\infty$.
We assume that $\operatorname{Argmin}(f + g) \neq \varnothing$. Let $x_0 \in \mathcal{H}$ and

$$(\forall n \in \mathbb{N}) \qquad \begin{cases} y_n = x_n - \gamma \nabla g(x_n) \\ x_{n+1} = x_n + \lambda_n(\operatorname{prox}_{\gamma f} y_n - x_n). \end{cases}$$

Then, $(x_n)_{n\in\mathbb{N}}$ converges weakly to a minimizer of $f + g$.

Optimization algorithms: *Forward-Backward*

Proof: Let $T = \mathrm{prox}_{\gamma f} \circ (\mathrm{Id} - \gamma \nabla g)$. For every $x \in \mathcal{H}$,

$$x \in \mathrm{Fix}\, T \Leftrightarrow (\mathrm{Id} - \gamma \nabla g)x \in (\mathrm{Id} + \gamma \partial f)x \Leftrightarrow 0 \in \nabla g(x) + \partial f(x).$$

Consequently, $\mathrm{Fix}\, T = \mathrm{zer}\,(\nabla g + \partial f) = \mathrm{zer}\,\big(\partial(g + f)\big) \neq \varnothing$.
Moreover, for every $n \in \mathbb{N}$,

$$x_{n+1} = x_n + \lambda_n (T x_n - x_n).$$

$\mathrm{prox}_{\gamma f}$ is 1/2-averaged and $\mathrm{Id} - \gamma \nabla g$ is $\gamma\nu/2$-averaged.
It follows that T is α-averaged with

$$\alpha = \frac{\frac{1}{2} + \frac{\gamma\nu}{2} - 2\frac{1}{2}\frac{\gamma\nu}{2}}{1 - \frac{1}{2}\frac{\gamma\nu}{2}} \quad \Leftrightarrow \quad \alpha^{-1} = \delta.$$

Optimization algorithms: *Forward-Backward* with varying stepsize

Let $f \in \Gamma_0(\mathcal{H})$.
Let $g \in \Gamma_0(\mathcal{H})$ be differentiable with a ν-Lipschitzian gradient where $\nu \in \,]0, +\infty[$.
Let $(\gamma_n)_{n\in\mathbb{N}}$ in $[\underline{\gamma}, \overline{\gamma}]$ where $0 < \underline{\gamma} < \overline{\gamma} < 2/\nu$ and
let $(\lambda_n)_{n\in\mathbb{N}}$ be a sequence in $[\underline{\lambda}, 1]$ with $0 < \underline{\lambda} \leq 1$.
We assume that $\operatorname{Argmin}(f+g) \neq \varnothing$. Let $x_0 \in \mathcal{H}$ and

$$(\forall n \in \mathbb{N}) \qquad \begin{cases} y_n = x_n - \gamma_n \nabla g(x_n) \\ x_{n+1} = x_n + \lambda_n(\operatorname{prox}_{\gamma_n f} y_n - x_n). \end{cases}$$

Then, $(x_n)_{n\in\mathbb{N}}$ converges weakly to a minimizer of $f+g$.

Optimization algorithms: projected gradient

Let C be a nonempty closed convex subset of $\mathcal{H}$.
Let $g \in \Gamma_0(\mathcal{H})$ be differentiable with a ν-Lipschitzian gradient where $\nu \in \left]0, +\infty\right[$.
Let $\gamma \in \left]0, 2/\nu\right[$ and $\delta = 2 - \gamma\nu/2 \in \left]1, 2\right[$.
Let $(\lambda_n)_{n\in\mathbb{N}}$ be a sequence in $[0, \delta]$ such that $\sum_{n\in\mathbb{N}} \lambda_n(\delta - \lambda_n) = +\infty$.
We assume that $\operatorname*{Argmin}_{x\in C} g(x) \neq \varnothing$. Let $x_0 \in \mathcal{H}$ and

$$(\forall n \in \mathbb{N}) \qquad \begin{cases} y_n = x_n - \gamma\nabla g(x_n) \\ x_{n+1} = x_n + \lambda_n(P_C y_n - x_n). \end{cases}$$

Then, $(x_n)_{n\in\mathbb{N}}$ converges weakly to a minimizer of g over C.

Optimization algorithms: gradient descent

Let $g \in \Gamma_0(\mathcal{H})$ be a differentiable function with a ν-Lipschitzian gradient where $\nu \in]0, +\infty[$.
Let $(\gamma_n)_{n\in\mathbb{N}}$ in $[\underline{\gamma}, \overline{\gamma}]$ where $0 < \underline{\gamma} < \overline{\gamma} < 2/\nu$.
We assume that $\operatorname{Argmin} g \neq \varnothing$. Let $x_0 \in \mathcal{H}$ and

$$(\forall n \in \mathbb{N}) \qquad x_{n+1} = x_n - \gamma_n \nabla g(x_n)$$

Then, $(x_n)_{n\in\mathbb{N}}$ converges weakly to a minimizer of g.

Optimization algorithms: proximal point algorithm

Let $f \in \Gamma_0(\mathcal{H})$.
Let $(\gamma_n)_{n\in\mathbb{N}}$ be a sequence in $]0,+\infty[$ such that $\sum_{n\in\mathbb{N}} \gamma_n = +\infty$. We assume that $\operatorname{Argmin} f \neq \varnothing$. Let $x_0 \in \mathcal{H}$ and

$$(\forall n \in \mathbb{N}) \qquad x_{n+1} = \operatorname{prox}_{\gamma_n f} x_n.$$

Then, $(x_n)_{n\in\mathbb{N}}$ converges weakly to a minimizer of f.

Optimization algorithms: Douglas-Rachford

Motivation

Let $f \in \Gamma_0(\mathcal{H})$ and $g \in \Gamma_0(\mathcal{H})$. We want to

$$\underset{x \in \mathcal{H}}{\text{minimize}} \;\; f(x) + g(x).$$

Possible solutions :

- gradient descent algorithm $\Rightarrow$ $f + g$ needs to be smooth
- proximal point algorithm $\Rightarrow$ $f + g$ needs to be "proximable"
- Forward-Backward algorithm $\Rightarrow$ g needs to be smooth

Can we find a splitting algorithm when both f and g are nonsmooth?

Optimization algorithms: Douglas-Rachford

Let $\gamma \in]0, +\infty[$ and let $f \in \Gamma_0(\mathcal{H})$.
The reflection of the proximal operator defined as

$$\mathrm{rprox}_{\gamma f} = 2\mathrm{prox}_{\gamma f} - \mathrm{Id}$$

is nonexpansive.

Optimization algorithms: Douglas-Rachford

Let $\gamma \in]0, +\infty[$ and let $f \in \Gamma_0(\mathcal{H})$.
The reflection of the proximal operator defined as

$$\mathrm{rprox}_{\gamma f} = 2\mathrm{prox}_{\gamma f} - \mathrm{Id}$$

is nonexpansive.

- Exercise: Establish this property.

Optimization algorithms: Douglas-Rachford

Let $\gamma \in]0, +\infty[$ and let $f \in \Gamma_0(\mathcal{H})$ and $g \in \Gamma_0(\mathcal{H})$.
We have

$$\operatorname{zer}(\partial f + \partial g) = \operatorname{prox}_{\gamma g}(\operatorname{Fix} T)$$

where $T = \operatorname{rprox}_{\gamma f} \circ \operatorname{rprox}_{\gamma g}$.

Optimization algorithms: Douglas-Rachford

Let $\gamma \in]0, +\infty[$ and let $f \in \Gamma_0(\mathcal{H})$ and $g \in \Gamma_0(\mathcal{H})$.
We have

$$\operatorname{zer}(\partial f + \partial g) = \operatorname{prox}_{\gamma g}(\operatorname{Fix} T)$$

where $T = \operatorname{rprox}_{\gamma f} \circ \operatorname{rprox}_{\gamma g}$.

Proof: Let $x \in \mathcal{H}$.

$$\begin{aligned}
0 \in \gamma\big(\partial f(x) + \partial g(x)\big) &\Leftrightarrow (\exists y \in \mathcal{H})\ \ x - y \in \gamma \partial f(x) \text{ and } y - x \in \gamma \partial g(x) \\
&\Leftrightarrow (\exists y \in \mathcal{H})\ \ 2x - y \in (\mathrm{Id} + \gamma \partial f)x \\
&\qquad\qquad \text{and } x = \operatorname{prox}_{\gamma g}(y) \\
&\Leftrightarrow (\exists y \in \mathcal{H})\ \ x = \operatorname{prox}_{\gamma f}\big(\operatorname{rprox}_{\gamma g}(y)\big) \text{ and } x = \operatorname{prox}_{\gamma g}(y) \\
&\Leftrightarrow (\exists y \in \mathcal{H})\ \ \operatorname{rprox}_{\gamma f}\big(\operatorname{rprox}_{\gamma g}(y)\big) = 2x - \operatorname{rprox}_{\gamma g}(y) = y \\
&\qquad\qquad \text{and } x = \operatorname{prox}_{\gamma g}(y) \\
&\Leftrightarrow (\exists y \in \operatorname{Fix} T)\ \ x = \operatorname{prox}_{\gamma g}(y).
\end{aligned}$$

Optimization algorithms: Douglas-Rachford

Let $\mathcal{H}$ be a Hilbert space.

Let $f \in \Gamma_0(\mathcal{H})$ and $g \in \Gamma_0(\mathcal{H})$.

$$(\forall n \in \mathbb{N}) \qquad \begin{cases} y_n = \operatorname{prox}_{\gamma g}(x_n) \\ z_n = \operatorname{prox}_{\gamma f}(2y_n - x_n) \\ x_{n+1} = x_n + \lambda_n(z_n - y_n). \end{cases}$$

Optimization algorithms: Douglas-Rachford

Let $\mathcal{H}$ be a Hilbert space.

Let $f \in \Gamma_0(\mathcal{H})$ and $g \in \Gamma_0(\mathcal{H})$.

Let $\gamma \in]0, +\infty[$ and let $(\lambda_n)_{n\in\mathbb{N}}$ be a sequence in $[0, 2]$ such that $\sum_{n\in\mathbb{N}} \lambda_n(2 - \lambda_n) = +\infty$.

We assume that $\operatorname{zer}(\partial f + \partial g) \neq \varnothing$. Let $x_0 \in \mathcal{H}$ and

$$(\forall n \in \mathbb{N}) \quad \begin{cases} y_n = \operatorname{prox}_{\gamma g}(x_n) \\ z_n = \operatorname{prox}_{\gamma f}(2y_n - x_n) \\ x_{n+1} = x_n + \lambda_n(z_n - y_n). \end{cases}$$

The following properties are satisfied:

- $x_n \rightharpoonup \widehat{x}$
- $z_n - y_n \to 0$, $y_n \rightharpoonup \widehat{y}$, $z_n \rightharpoonup \widehat{y}$ where $\widehat{y} = \operatorname{prox}_{\gamma g}\widehat{x} \in \operatorname{Argmin}(f + g)$.

Optimization algorithms: Douglas-Rachford

Proof: For simplicity, assume that $\mathcal{H}$ is finite dimensional.
Let $T = \operatorname{rprox}_{\gamma f} \circ \operatorname{rprox}_{\gamma g}$. T is nonexpansive and $\varnothing \neq \operatorname{zer}(\partial f + \partial g) = \operatorname{prox}_{\gamma g}(\operatorname{Fix} T) \Rightarrow \operatorname{Fix} T \neq \varnothing$. Moreover, for every $n \in \mathbb{N}$,

$$
\begin{aligned}
x_{n+1} &= x_n + \lambda_n\big(\operatorname{prox}_{\gamma f}(2\operatorname{prox}_{\gamma g}(x_n) - x_n) - \operatorname{prox}_{\gamma g}(x_n)\big) \\
&= x_n + \frac{\lambda_n}{2}\big(2\operatorname{prox}_{\gamma f}(2\operatorname{prox}_{\gamma g}(x_n - x_n) - 2\operatorname{prox}_{\gamma g}(x_n) + x_n - x_n\big) \\
&= x_n + \frac{\lambda_n}{2}\big(2\operatorname{prox}_{\gamma f}(\operatorname{rprox}_{\gamma g}(x_n)) - \operatorname{rprox}_{\gamma g}(x_n) - x_n\big) \\
&= x_n + \frac{\lambda_n}{2}\big(Tx_n - x_n\big).
\end{aligned}
$$

$\Rightarrow$ Krasnosel'skii-Mann algorithm with relaxation factors $(\lambda_n/2)_{n\in\mathbb{N}}$.
We deduce that $Tx_n - x_n \to 0$ and $x_n \to \widehat{x} \in \operatorname{Fix} T$.

Optimization algorithms: Douglas-Rachford

Proof: For every $n \in \mathbb{N}$,

$$z_n - y_n = \mathrm{prox}_{\gamma f}(2\mathrm{prox}_{\gamma g}(x_n) - x_n) - \mathrm{prox}_{\gamma g}(x_n) = \frac{1}{2}(Tx_n - x_n) \to 0.$$

Moreover, $\mathrm{prox}_{\gamma g}$ being continuous (since nonexpansive), we have $y_n \to \mathrm{prox}_{\gamma g}\widehat{x} \in \mathrm{zer}\,(\partial f + \partial g)$.
Because $z_n - y_n \to 0$, we have: $z_n \to \mathrm{prox}_{\gamma g}\widehat{x} \in \mathrm{zer}\,(\partial f + \partial g)$.
In addition,

$$\begin{aligned} & \varnothing \neq \partial f + \partial g \subset \partial(f+g) \\ \Rightarrow \quad & \widehat{y} = \mathrm{prox}_{\gamma g}(\widehat{x}) \in \mathrm{zer}\left(\partial(f+g)\right) = \mathrm{Argmin}(f+g). \end{aligned}$$

Optimization algorithms: Parallel form of Douglas-Rachford

Let $\mathcal{H}$ and $\mathcal{G}$ be two Hilbert spaces.
Let $g \in \Gamma_0(\mathcal{H})$ and $L \in \mathcal{B}(\mathcal{G}, \mathcal{H})$ be such that $\operatorname{ran} L$ is closed and L^*L is an isomorphism.
Let $\gamma \in \left]0, +\infty\right[$ and let $(\lambda_n)_{n\in\mathbb{N}}$ a sequence in $[0, 2]$ such that $\sum_{n\in\mathbb{N}} \lambda_n(2 - \lambda_n) = +\infty$.
We assume that $\operatorname{zer}(L^* \circ \partial g \circ L) \neq \varnothing$. Let $x_0 \in \mathcal{H}$, $v_0 = (L^*L)^{-1}L^*x_0$ and

$$(\forall n \in \mathbb{N}) \quad \begin{cases} y_n = \operatorname{prox}_{\gamma g}(x_n) \\ c_n = (L^*L)^{-1}L^*y_n \\ x_{n+1} = x_n + \lambda_n\big(L(2c_n - v_n) - y_n\big) \\ v_{n+1} = v_n + \lambda_n(c_n - v_n). \end{cases}$$

We have then
$v_n \rightharpoonup \widehat{v}$ where $\widehat{v} \in \operatorname{Argmin}(g \circ L)$.

Optimization algorithms: Parallel form of Douglas-Rachford

Sketch of proof:

$$\underset{v\in\mathcal{G}}{\text{minimize}}\ \ g(Lv) \quad\Leftrightarrow\quad \underset{x\in\mathcal{H}}{\text{minimize}}\ \ \iota_E(x)+g(x)$$

where $E = \operatorname{ran} L$.
We apply Douglas-Rachford algorithm with
$f = \iota_E \Rightarrow \operatorname{prox}_{\gamma f} = P_E$ by setting

$$(\forall n\in\mathbb{N}) \qquad P_E y_n = Lc_n \ \text{ and } \ P_E x_n = Lv_n$$

where $c_n = \underset{c\in\mathcal{H}}{\operatorname{argmin}}\ \|y_n - Lc\|^2 = (L^*L)^{-1}L^*y_n$.

Optimization algorithms: Parallel form of DR

Particular case of Douglas-Rachford algorithm:

$\mathcal{H} = \mathcal{H}_1 \times \cdots \times \mathcal{H}_m$ where $\mathcal{H}_1, \ldots, \mathcal{H}_m$ Hilbert spaces

$\big(\forall x = (x_1, \ldots, x_m) \in \mathcal{H}\big)\ g(x) = \sum_{i=1}^{m} g_i(x_i)$

where $(\forall i \in \{1, \ldots, m\})\ g_i \in \Gamma_0(\mathcal{H}_i)$

$L\colon v \mapsto (L_1 v, \ldots, L_m v)$ where $(\forall i \in \{1, \ldots, m\})\ L_i \in \mathcal{B}(\mathcal{G}, \mathcal{H}_i)$.

PPXA+ algorithm

Let $(x_{0,i})_{1 \leq i \leq m} \in \mathcal{H}$, $v_0 = \big(\sum_{i=1}^{m} L_i^* L_i\big)^{-1} \sum_{i=1}^{m} L_i^* x_{0,i}$ and

$$(\forall n \in \mathbb{N}) \quad \begin{cases} y_{n,i} = \operatorname{prox}_{\gamma g_i}(x_{n,i}), \quad i \in \{1, \ldots, m\} \\ c_n = \big(\sum_{i=1}^{m} L_i^* L_i\big)^{-1} \sum_{i=1}^{m} L_i^* y_{n,i} \\ x_{n+1,i} = x_{n,i} + \lambda_n\big(L_i(2c_n - v_n) - y_{n,i}\big), \quad i \in \{1, \ldots, m\} \\ v_{n+1} = v_n + \lambda_n(c_n - v_n). \end{cases}$$

We have then $v_n \rightharpoonup \widehat{v} \in \operatorname{Argmin} \sum_{i=1}^{m} g_i \circ L_i$.

Exercise

Let $m \in \mathbb{N}$, $m \geq 2$.

1. For every $i \in \{1, \ldots, m\}$, let g_i be a function in $\Gamma_0(\mathcal{H})$. Propose a parallel algorithm for finding a minimizer of $\sum_{i=1}^{m} g_i$ when such a minimizer exists.
2. Let $p \in \{1, \ldots, m-1\}$ and let $(C_i)_{1 \leq i \leq m}$ be nonempty closed convex subsets of $\mathcal{H}$. Deduce a parallel algorithm for solving the problem

$$\underset{x \in \bigcap_{i=1}^{p} C_i}{\text{minimize}} \sum_{j=p+1}^{m} \sigma_{C_j}(x)$$

when this problem admits a solution.